

Persistence in factor-based supervised learning models

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Abstract

In this paper, we document the importance of *memory* in machine learning (ML)-based models relying on firm characteristics for asset pricing. We find that predictive algorithms perform best when they are trained on long samples, with long-term returns as dependent variables. In addition, we report that persistent features play a prominent role in these models. When applied to portfolio choice, we find that investors are always better off predicting *annual* returns, even when rebalancing at lower frequencies (monthly or quarterly). Our results remain robust to transaction costs and risk scaling, thus providing useful indications to quantitative asset managers.

Keywords: Factor investing, Machine learning, Asset Pricing, Autocorrelation

JEL codes: C45, C53, G11, G12

1. Introduction

The academic asset pricing literature has seen a recent surge in contributions resorting to machine learning (ML) methods to estimate models or perform predictions (see, e.g., [Chen, Pelger, and Zhu \(2019\)](#), [Gu, Kelly, and Xiu \(2021\)](#), [Gu, Kelly, and Xiu \(2020\)](#), and [Feng, Polson, and Xu \(2019\)](#)). Because these sophisticated tools seem to deliver higher predictive accuracy, it is tempting to use them for portfolio allocation purposes, as is for instance done in [Krauss, Do, and Huck \(2017\)](#), [Aldridge and Avellaneda \(2019\)](#) and [Gu et al. \(2020\)](#).

In supervised learning, the key question is whether a model fitted on a given database (the training set) will perform well on previously unseen data (the testing set). Models that manage to uphold performance out-of-sample are said to *generalize* well because their predictive ability is not limited to in-sample tests. In financial economics, one key attribute of training and testing sets is their natural chronological ordering: models are trained on data prior to some time t and tested on realizations posterior to t . This sequential process is illustrated in Figure 1.

The key question is thus: will what is learned from the past hold in the future? In this paper, we argue that models are more likely to generalize well when they *memorize*, that is, when the patterns they detect do not rapidly vanish. In practice, *enduring patterns* are hard to define and quantify, thus it is natural to resort to simpler concepts. The econometric literature has one such notion that synthetically encapsulates the key idea of memory: autocorrelation.

An ideal framework would allow to evaluate the autocorrelation of a *model*. Unfortunately, there is no unambiguous way to compute such a quantity because models can be very complex objects. It is much easier to obtain the autocorrelograms both of the series that are used to train the models (labels and features) and of the errors stemming from the model. In this paper,

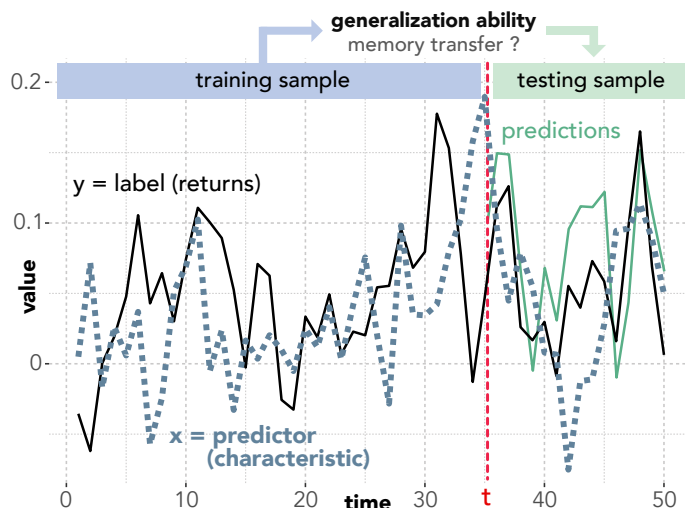


Fig. 1. **Illustration of the concept of chronological generalization for machine learning models.** The model learns from a past sample, but its performance is assessed over future dates. If the accuracy is sustained out-of-sample, the model is said to generalize well.

we argue that the performance of the models is strongly driven by the autocorrelations of the inputs that are used to calibrate (i.e., train) the models.

In the traditional literature on statistical inference for linear models, persistence in dependent and independent variables is troublesome. When regressors are persistent, estimators are biased but can be corrected (Stambaugh (1999), Amihud and Hurvich (2004) and Hjalmarsson (2008b)), while statistical tests must also be improved (Campbell and Yogo (2006), Chen and Deo (2009)). When the dependent variable (often, returns) is computed over several periods and is therefore autocorrelated as well, other issues arise, which can also be tackled econometrically (Hjalmarsson (2011), Xu (2020)). In fact, predictive regressions relying on long term returns have a documented tendency to generate spurious significance (Campbell (2001), Valkanov (2003), Boudoukh, Richardson, and Whitelaw (2008) and Phillips and Lee (2013)).

From a purely predictive point of view, these issues matter much less. Inference results, while insightful, are not central to portfolio backtests for which the only yardstick that matters is pure out-of-sample accuracy. In this paper, we argue that having autocorrelated series on both sides of model equations helps increase their generalization abilities. Indeed, working with processes that exhibit memory increases the odds of injecting part of this memory in the models so that the patterns that are observed in-sample endure out-of-sample. Heuristically, these models are more able to identify lasting trends and less prone to detect short-term fads. The rationale for our results is thus simple. Series that exhibit ever-changing patterns are harder to forecast. Some kind of stability is required to improve out-of-sample precision.¹ We argue that persistence, evaluated as sample autocorrelation, is a fairly good proxy of stability.

Our contributions start with a simple theoretical toy model that helps grasp one concept at stake in the paper. The key result shows that when autocorrelation is larger than some proxy for noise, the concentration of signal is larger for two-period returns than for their one period

¹In the machine learning literature, some authors talk about *invariance*, as in Arjovsky, Bottou, Gulrajani, and Lopez-Paz (2019). This article notably shows that invariance can be related to causality, which is a key concept in statistical learning. Simply put, and paraphrasing Jeff Bezos: being able to guess what *does not change* is more useful than being able to conjecture what *will change*.

counterpart: the informational content diffuses and is compounded through time. In addition, our main empirical findings are fourfold. First, we report that the pure out-of-sample fit of the models can be disappointing, but that it may also be irrelevant. Indeed, most of the fit is driven by the variations of the dependent variable, which have little pertinence. The useful term, the correlation between predicted returns and realized returns is drowned in the noise of their variances. When this correlation is well above zero on average, the cross-sectional rankings of stock returns are relatively accurate, which generates profits. As in Lettau and Pelger (2020a) and Lettau and Pelger (2020b), we argue that focusing more on cross-sectional fit and less on global fit is beneficial to out-of-sample portfolio allocation. Our second core finding is that accuracies (and profits) are the largest when the predictive models are built on deep samples and when the dependent variable is the long-term future return. Third, we document that important drivers of returns depend on the horizon of the returns. Liquidity and momentum matter in the short term while fundamentals are more prominent for long term returns. Finally, our conclusions regarding portfolio choice are robust to transaction costs.

The paper is structured as follows. In Section 2, we document the importance of persistence with respect to the fit of supervised learning models. Applications of our baseline results to portfolio choice are discussed in Section 3. Section 4 is dedicated to extensions. Notably, we investigate the relative importance of predictor variables and its relationship to persistence. Finally, we conclude in Section 5. For the sake of transparency, the material used for the paper (data and code) is available online.²

2. Return horizon and the cross-sections of autocorrelations

In financial econometrics, monthly returns are weakly autocorrelated (see Campbell, Lo, and MacKinlay (1997)). Computing returns at longer horizons (e.g., quarterly, or annually) mechanically increases autocorrelation because of the resulting overlaps. If p_t is the price of an asset at month t , then two consecutive annual returns $p_t/p_{t-12} - 1$ and $p_{t+1}/p_{t-11} - 1$ have an overlap of 11 months.

From an inferential standpoint, the statistical significance of predictors has widely been documented to increase with the horizon of the returns.³ Whether or not this translates into increased out-of-sample accuracy remains an open question because increasing return horizon shifts training samples back in time. Indeed, in order to predict annual returns at time t , the training sample must end at $t - 12$ to prevent any forward-looking leakage. Any annual return computed posterior to $t - 12$ assumes knowledge of prices beyond t . Thus, the design of training samples requires extra care.

The purpose of this section is to examine the impact of return horizon on predictive accuracy. We start with a stylized toy framework that exposes the principle for simple linear models. We then move on to a detailed empirical analysis with a subsequent focus on portfolio decisions.

²It can be accessed at <http://www.gcoqueret.com/horizons.html>.

³See, e.g. Fama and French (1988b,a). The phenomenon seems to vanish for very long horizons beyond 12 to 15 years (Bandi, Perron, Tamoni, and Tebaldi (2019)). In fact, it has been shown that longer horizons distort inference (Valkanov (2003), Boudoukh et al. (2008), Hjalmarsen (2008a)), which is why researchers have proposed technical solutions to improve estimators and tests (see Hjalmarsen (2011), Phillips and Lee (2013), Sizova (2014), Chevillon (2017), Xu (2020) and Boudoukh, Israel, and Richardson (2020)).

2.1. Signal intensity in predictive regressions

We begin by providing some intuition via a stylized toy model. The simplest forecasting equation for returns can be written as follows:

$$r_{t+1} = \beta C_t + \epsilon_{t+1}^r, \quad \beta \neq 0,$$

where C_t is the lagged predictor value and ϵ_{t+1}^r is the error term.⁴ We assume that C_t follows an AR(1) process with autocorrelation ρ_C and variance of innovations σ_C^2 , so that its unconditional variance is $\sigma_C^2(1 - \rho_C^2)^{-1}$. Likewise, the variance of errors is denoted with σ_ϵ^2 .

In this setup, we are interested in the signal-to-noise (SN) ratio, which we define as

$$\text{SN}(r_t) = \frac{\mathbb{V}[\beta C_t]}{\mathbb{V}[\epsilon_t]} = \beta^2 \frac{\sigma_C^2}{\sigma_\epsilon^2(1 - \rho_C^2)}.$$

The question we then ask is: how does the SN ratio of long-term returns compare with that of r_t . To simplify the computations, we restrict the analysis to the 2 period ahead return:

$$r_* = (1 + r_{t+1})(1 + r_{t+2}) - 1. \quad (1)$$

The proposition below describes the condition (necessary and sufficient) under which the SN ratio of the two-period return r_* is larger than that of r_t . The proof can be found in [Appendix C](#).

Proposition 1. *Assuming $\beta \neq 0$,*

$$\text{SN}(r_*) \geq \text{SN}(r_t) \iff \frac{\sigma_\epsilon^2 + \beta^2 \sigma_C^2}{2} \leq \rho_C. \quad (2)$$

This result is not intuitive at first sight. It implies that if a particular linear combination of the variances in the model is smaller than the autocorrelation of the characteristic (predictor), then the return computed over two period has a larger signal-to-noise ratio. In finance, the variances may well be small in magnitude, which means that for highly autocorrelated features, the condition can be verified as long as β is not too large. In this case, it is beneficial to predict longer term returns instead of one-period ahead returns because the agent is able to extract more signal thereof. This idea is hardly new and was already highlighted in [Valkanov \(2003\)](#): “*intuitively, the aggregation of a series into a long-horizon variable is thought to strengthen the signal, while eliminating the noise.*”

2.2. Lessons from simple simulations

To further illustrate our intention, let us consider a new framework with 2 independent processes: a feature C_t which is either mildly persistence ($\rho_C = 0.5$) or highly autocorrelated ($\rho_C = 0.95$) and a return r_t^ρ which is an AR(1) process with autocorrelation ρ . We seek to compare the relevance of the following linear models:

$$r_t^\rho = a^\rho + b^\rho C_t + e_t \quad (3)$$

⁴This specification is the most common case of (linear) predictive regressions. We refer to [Stambaugh \(1999\)](#) and more recently to the survey of [Gonzalo and Pitarakis \(2018\)](#) for technical details on estimation techniques and related issues.

for different value of ρ . The above relationship is not predictive but it does not matter since both processes are independent (lagging the regressor would yield the same results). In addition, there is no causal relationship between r and C in the generation of the data. This specification may seem odd, thus we briefly comment on it. In traditional inference, the econometrician would start by assuming that the relationship (3) holds, and then estimate its coefficients and conclude based on distributional properties of the errors. Here, the purpose is different: we fix the data generating process and evaluate the out-of-sample accuracy of the OLS estimator. Naturally, we could assume some correlation between r and C , but it is shown in Figures 3 and 4 of Coqueret and Deguest (2020), that this would only marginally impact the outcome.

The rolling OLS estimates depend on time and the sample depth, D : $\hat{a}_{t,D}^\rho$ and $\hat{b}_{t,D}^\rho$. We roll the models in time with fixed windows of size D and we predict the one step ahead return with these dynamic models. We are then able to compute the mean squared error:

$$\text{MSE}(\rho, D) = \frac{1}{T} \sum_{t=1}^T (r_t^\rho - \hat{a}_{t,D}^\rho - \hat{b}_{t,D}^\rho C_t)^2. \quad (4)$$

In Figure 2, we plot the values of MSE for a range of depths and for four levels of autocorrelations of returns ($\rho = 0.05, 0.5, 0.85, 0.95$). We drew 10^4 samples for each process with standard deviations $\sigma_r = 0.2$ and $\sigma_C = 0.05$. This implies that the variance of *innovations* in the AR(1) return process is dependent on the autocorrelation level: $\sigma_e = \sigma_r \sqrt{1 - \rho^2}$.

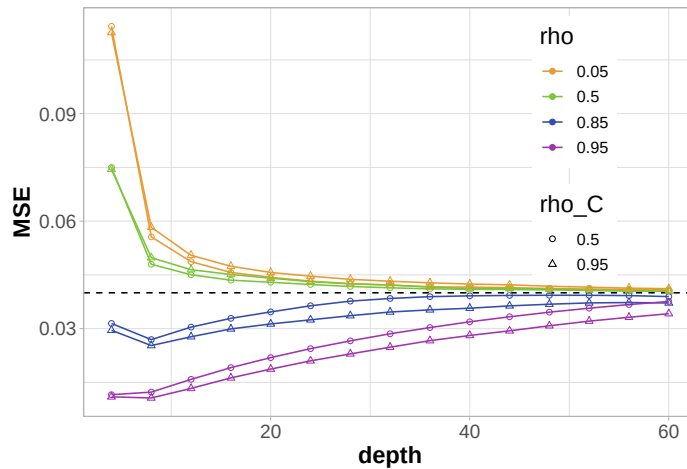


Fig. 2. **Mean squared errors of predictions with autocorrelated returns.** We plot the MSE defined in Equation (4) for four series of AR(1) returns characterized by their level of autocorrelation ρ (shown with colors): 0.05, 0.5, 0.85 and 0.95. All four processes for returns have the same standard deviation, $\sigma = 0.2$. The two levels of persistences for the predictors are depicted with shapes (circles for $\rho_C = 0.5$ and triangles for $\rho_C = 0.95$). The total number of simulated points for each process is 10^4 . The asymptotic limit of the MSE when sample depth increases to infinity is $\sigma_r^2 = 0.04$ and it is shown with a dashed black line.

We make two important observations. First, the out-of-sample error decreases with the autocorrelation of the return: there is a clear ordering of colors, though it becomes more blurry when the depth increases. Given the fact that the two processes are independent, these stylized properties may be spurious, but this notion is out of the scope of the present paper and investigated in a separate article.

Second, the persistence of the predictors does not matter much (the two shapes (circle and triangle) are close, which generates twin lines for each value of ρ). This points to a rather obvious conclusion: absent any causality relationship, it is easier to forecast persistent time-series. By construction, persistence in returns is obtained by increasing the horizon at which they are computed. This is an important point because any predictive model that aims to forecast annual returns cannot be trained on predictors that have been released in the past year. This would indeed generate a look ahead bias in the estimation. Thus, a model based on annual returns is trained on data that may be slightly out-dated. From these two opposing effects (gain from persistence but loss from data obsolescence), it is hard a priori to know which is the strongest, which is why we next turn to our empirical study.

2.3. Data

The dataset is the same as the one used in the book by Coqueret and Guida (2020), which is an aggregation of fields from numerous data vendors.⁵ It comprises fundamental data for 1,206 large firms listed in North America (US overwhelmingly).⁶ The sample spans the years 2000 to 2019. To make our backtest more realistic, we refrain from any survivorship bias and keep firms which have been acquired or have bankrupt. The total number of firms in the sample is thus not constant.

Each stock is characterized by 93 characteristics⁷ listed in Tables 2 to 4 in the Appendix. These variables, sampled at the monthly frequency, and which we use as predictors, are documented drivers of the cross-section of stock returns (e.g., accounting figures and ratios, risk and liquidity measures or momentum trends). When expressed in currency units, all numbers are converted to US dollars.

Most predictors depend on fundamental data that is disclosed every quarter (e.g., earnings and dividends). We process the series so that any missing value is replaced by the last non-missing value, unless there are more than 12 consecutive missing points (more than one year of data), in which case the observation is discarded. Thus, some fields are constant over periods of three months. Other fields rely on accounting *and* market data, like the price-to-book ratio, which is thus updated on a monthly basis because prices are available on a daily basis. Nevertheless, because prices and market capitalization are relatively slow moving, this implies that such predictors are, too, rather persistent.

The features are moreover uniformized so that at each date, each predictor is uniformly distributed (across stocks) over the unit interval. Scaling predictors is a widespread practice both in the machine learning community and in some recent asset pricing articles (see, e.g., Kojien and Yogo (2019), Kelly, Pruitt, and Su (2019) and Freyberger, Neuhierl, and Weber (2020)). We work with four horizons for future returns which serve as dependent variables in the models: 1, 3, 6 and 12 months. Annual returns, when observed at a monthly frequency, are sufficiently autocorrelated: there is no need to resort to longer horizons. Moreover, holding periods in the factor investing industry seldom exceed one year.

For each stock and each characteristic, one key indicator is the first order sample autocorrelation value. In Figure 3, we plot the distribution of these values across all assets and features.

⁵The book is accessible online at www.mlfactor.com. The material (e.g., the dataset) can be obtained at <https://github.com/shokru/mlfactor.github.io/tree/master/material>.

⁶The sample comprises firms that are sufficiently liquid and corresponds to a typical investment universe of an international asset manager.

⁷In the sequel, we use the terms *characteristics*, *indicators*, *features* and *predictors* interchangeably.

The median sample autocorrelation of features is 0.835. For each predictor, the average autocorrelation across all stocks is also reported in the tables in the Appendix. The high persistence in these variables can either come from inherent smoothness (e.g., market capitalization is stable both in time and in the cross-section), or by construction. For instance, long-term volatility from one month to another is computed over large overlapping sample, which naturally generates autocorrelation. Likewise, accounting figures, such as earnings or dividends are only released at quarterly frequencies, so that constant values may also generate possibly spurious persistence.

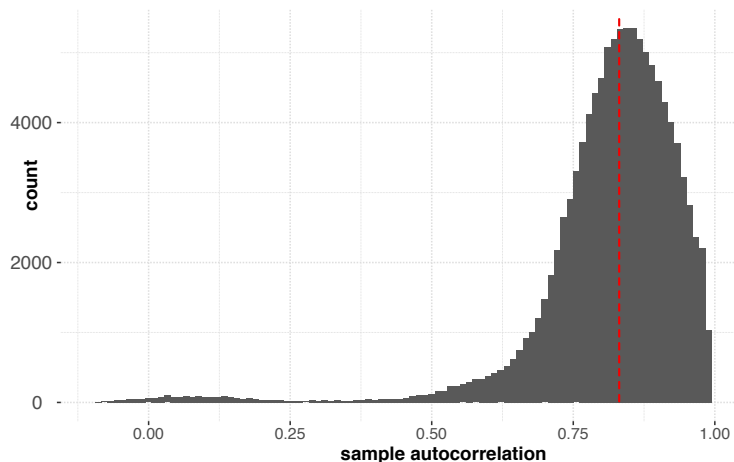


Fig. 3. Distribution of first order autocorrelation. We plot the histogram of the first autocorrelation $\mathbb{E}[(x_{t,n}^{(k)} - \bar{x}_n^{(k)})(x_{t+1,n}^{(k)} - \bar{x}_n^{(k)})]$ for all stocks n and their characteristics k . The vertical red dashed line marks the median of the distribution.

2.4. Protocol for supervised learning algorithms

For simplicity, we test two algorithms that belong to classes which have been documented to perform well in many portfolio allocation exercises (Krauss et al. (2017), Gu et al. (2020)). The first algorithm belongs to the family of boosted trees. Boosted trees aggregate simple decision trees and each new tree added to the ensemble aims to improve the fit of the model. We refer to Chen and Guestrin (2016) for a detailed account of the implementation of this learning engine.

These models can be very flexible and may depend on a large array of parameters. Three of them stand out as particularly important: the number of trees used, the learning rate (which acts as a discount factor applied to each new tree added to the model) and the maximum depth of the trees that are grown. We choose common hyperparameter values because we do not want to overfit through validation optimization. Indeed, the goal of the paper is not to find the ultimately optimal models but to assess the impact of the design of the training set on the quality of the outcome. Working with fixed hyperparameters also makes our results more reproducible.

Consequently, we work with 80 trees, with 4 level trees and a learning rate of $\eta = 0.2$.⁸ These two parameters determine the complexity of the algorithm. Four level trees imply $\sum_{n=0}^3 2^n = 15$ splits, each with two decisions (splitting variable and splitting level), so that the maximum total number of trained parameters for 80 trees is 2,400. While this may seem a lot, it is much lower than the number of observations for six month samples (which are close to 86,400).

⁸The other parameters are the ones set by default in the XGBoost library, see <https://xgboost.readthedocs.io>.

The second algorithm is a feed-forward neural network (NN). We refer to Chapter 4 of [Du and Swamy \(2013\)](#) for a theoretical introduction on the subject of NNs and to Chapter 8 in [Coqueret and Guida \(2020\)](#) for their applications in factor investing. The network has two hidden layers, associated with rectified linear unit (ReLU) and hyperbolic tangent activation functions. The latter are favorable because the range of their output $[-1,1]$ is suitable for returns (see [Aldridge and Avellaneda \(2019\)](#)). The output layer has no activation. We keep a low dimensional network on purpose: [Gu et al. \(2020\)](#) have shown that deep architectures do not increase the predictive power and are often detrimental to accuracy. Deeper models have more parameters and are associated with higher odds of overfitting. The first layer of the networks has 16 units while the second has 8. Given the fact that the sample encompasses 93 predictors, this implies that there are 1,649 parameters in total that are trained (this figure is comparable to that of the boosted tree model).⁹ The learning process operates simple backpropagation rounds on batches of size 512 and over 10 epochs. Increasing the training, e.g., to 20 epochs, does not improve out-of-sample fit. These implementation choices are commonplace and have little impact on our conclusions.¹⁰ For both algorithms, the loss function is the sum of squared errors: we perform regression tasks.

In order to rigorously describe our empirical setting, we deploy a formal set of notations. The full machine learning formulation of the problem is the following. The aim is to estimate (learn) functions f that approximate future returns:¹¹

$$r_{t:t+\Delta T,i} = f(\mathbf{x}_{t,i}) + \epsilon_{t,i,\Delta T}, \quad (5)$$

where the vector $\mathbf{x}_{t,i}$ stacks the characteristics' values for stock i at time t and serves as feature for the two predictive families f described in the previous section. The function f is common to all assets, hence the framework is closer to panel models than to traditional predictive regressions for which each asset has its own model and sets of parameters. A temporal view of the sampling is outlined in Figure 4.

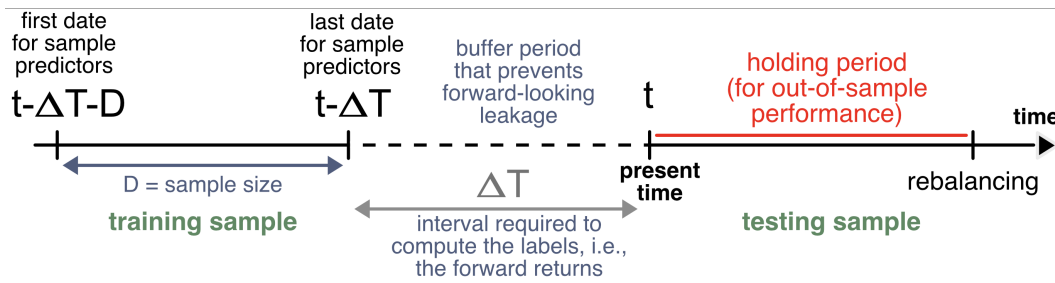


Fig. 4. Sampling diagram.

The models f are updated (re-trained) dynamically at each period t . To each period t is assigned a training set $\mathcal{S}_{t-1}(\Delta T, D)$ which compiles the data (label and features) for all date-firm

⁹Baum and Haussler (1989) provide a theoretical **linear** link between the number of samples and the network complexity, but the multiplicative constant depends on the data generating process and on the number of nodes. Bartlett (1998) argues that the magnitude of weights may also be an important driver of neural networks' accuracy. A common rule of thumb is that the number of samples should be larger than the number of weights in the network. While this can be debated (see, e.g., Lawrence, Giles, and Tsoi (1998)), we stand by this assumption in this paper.

¹⁰Results for smaller or more complex models are available upon request.

¹¹Following the machine learning jargon, we sometimes call these returns the 'labels', or simply, the dependent variable.

couples (s, j) prior (strictly) to t . These training sets critically depend on two numbers: on ΔT because they are naturally driven by the maturity of the label and on D , which is the depth of the training sample, i.e., how far back in the past it reaches. Lastly, the models f are also indexed by their family, M , which can take only two values (one for boosted trees, the other for neural networks). At time t , the models f are trained on the sets $\mathcal{S}_{t-1}(\Delta T, D)$ so that the sum of squared errors is minimized (at least locally). The predictions are then written

$$\hat{f}_t^M(\mathbf{x}_{t,i}, \mathcal{S}_{t-1}(\Delta T, D)), \quad (6)$$

which shows their dependence to the *past* training sample $\mathcal{S}_{t-1}(\Delta T, D)$ and to the *current* characteristic values $\mathbf{x}_{t,i}$.

We test four different values for ΔT : 1, 3, 6 and 12 months, which are the most common rebalancing frequencies in the factor investing industry. The sample depths are $D \in \{6, 12, 24, 48, 96\}$ months and will allow both very reactive or quite stable models. In addition, we run a backtest that starts with 96 months of training data and then expands the training window each time we update the model, that is, each month. Expanding windows are for instance used in [Gu et al. \(2020\)](#).

The evaluation starts in January 2009 because we need $8+1=9$ years of buffer for the first training sample. The additional year in the buffer period comes from the maximum maturity of 12 months for the label and is imperative to prevent any forward looking leakage. In total, over all possible combinations of dates, model types, depths and labels, we trained 5,856 models and made 7.07 million return predictions.

2.5. Baseline results

As is customary in the forecasting literature, we start by measuring the global fit of the predictions. We borrow the definition of the out-of-sample R^2 from [Gu et al. \(2020\)](#):

$$R_{\text{OOS}}^2(\Delta T, D, M) = 1 - \frac{\sum_{(t,i) \in \mathbb{S}_{\text{OOS}}} \left(r_{t:t+\Delta T,i} - \hat{f}_t^M(\mathbf{x}_{t,i}, \mathcal{S}_{t-1}(\Delta T, D)) \right)^2}{\sum_{(t,i) \in \mathbb{S}_{\text{OOS}}} r_{t:t+\Delta T,i}^2}, \quad (7)$$

where $\hat{f}_t^M$ is the estimated model and $\mathbb{S}_{\text{OOS}}$ is the testing sample from 2009 onwards. We purposefully use very different notations between the sets $\mathbb{S}_{\text{OOS}}$ and $\mathcal{S}_t$: the former is a fixed set of dates and assets while the latter is dynamically rolled as the backtest covers all dates of $\mathbb{S}_{\text{OOS}}$.

The only difference between Equation (7) and a classical R^2 measure is that the denominator of the second term is not the (scaled) variance of the dependent variable, but the sum of its squared values, i.e., there is no demeaning. As is discussed in [Gu et al. \(2020\)](#), this specification is *more conservative*, as it reduces the values of the R^2 .

In Figure 5, we report the values of $R_{\text{OOS}}^2(\Delta T, D)$ for the chosen levels of sample sizes D and label maturities ΔT . The most striking observation is that most figures are negative. Essentially, this means that the quadratic fit of the forecasts is lower than that of an agnostic model that predicts zero returns for all assets at all dates. [In unreported results, we tested other parametric configurations when tuning the complexity of the algorithms. The results remained qualitatively unaltered, meaning that our conclusion are not contingent on the choice of hyperparameters.](#)

A second conclusion is that the out-of-sample fit increases with sample sizes (D) but is hardly impacted by label maturities (ΔT). Overall, as the sample size increases, the magnitude

of errors shrinks. Finally, a third pattern is the apparent discrepancy between the two model types: neural networks are associated to more favorable (less negative) measures compared to boosted trees.

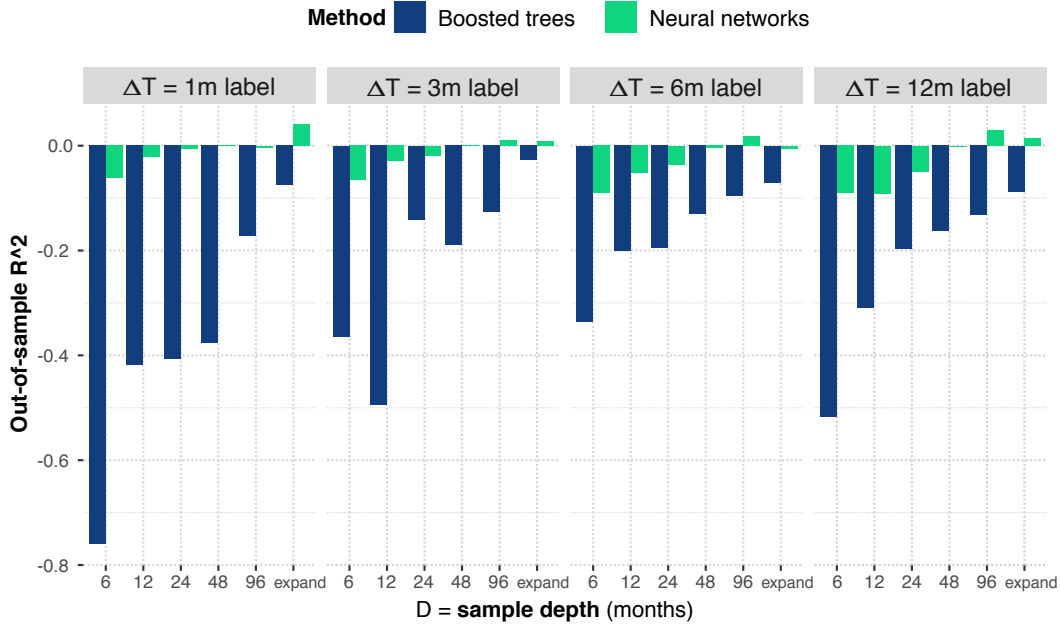


Fig. 5. **Global fit.** We plot the R_{oos}^2 metric for the two model types (boosted trees and neural networks). Predictions are formed between January 2009 and February 2019 on samples of size D . The dependent variable is the future return with maturity ΔT .

To further test the quantitative discrepancy between the two model types, we compute the modified [Diebold and Mariano \(1995\)](#) test statistic proposed in [Gu et al. \(2020\)](#). It is calculated as

$$\text{DM}_t(\Delta T, D) = \frac{\bar{d}_t(\Delta T, D)}{\hat{\sigma}(d_t(\Delta T, D))}; \quad (8)$$

$$d_t(\Delta T, D) = \frac{1}{N_t} \sum_{i=1}^{N_t} (\hat{\epsilon}_{t,i,\Delta T,D,\text{BT}}^2 - \hat{\epsilon}_{t,i,\Delta T,D,\text{NN}}^2), \quad (9)$$

where N_t is the number of stocks available at time t . In Equation (8), $\bar{x}$ and $\hat{\sigma}(x)$ refer to the mean and standard deviation of x . In Equation (9), the subscripts in errors refer to **B**oosted **T**rees (BT) and **N**eural **N**etworks (NN). The series d_t compute the average difference between squared errors of predictions. We plot the statistics in Figure 6 below. Their values oscillate through time and are more stable when the label has a longer maturity, though this conclusion is less salient for the short term label. The main lesson from Figure 6 is nonetheless that the statistics are small in magnitude and inside intervals of any reasonable confidence threshold. Therefore, it is unlikely that one type of algorithms has a significant edge over the other, which contradicts the seeming contrast observed in Figure 5.

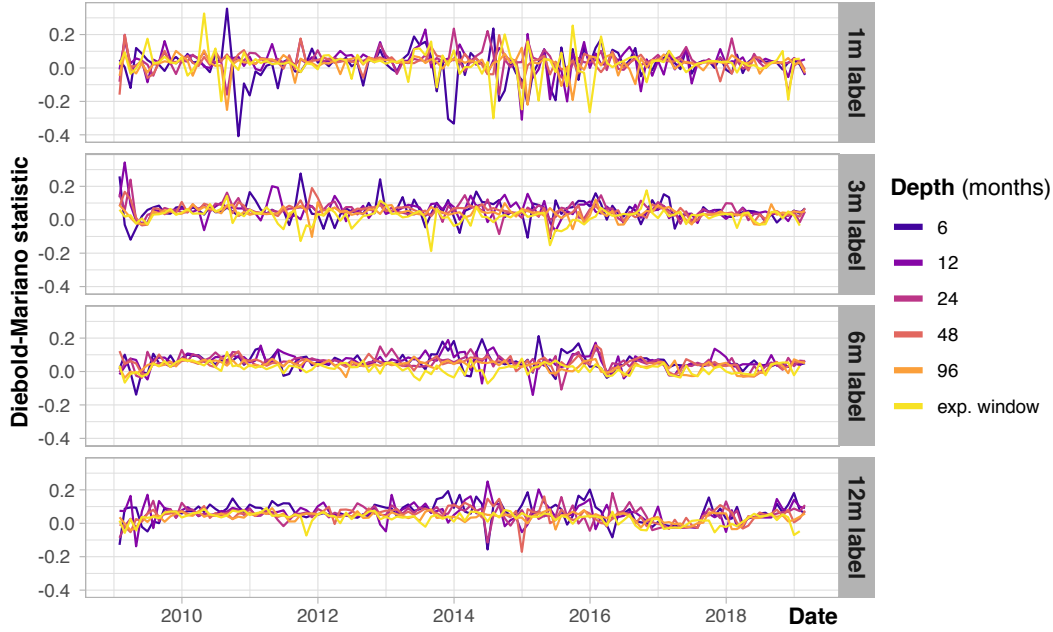


Fig. 6. **Diebold-Mariano statistics.** We plot the DM statistics as a function of time, training depth D , and label maturity ΔT .

3. Implications for portfolio choice

3.1. Simple sorts

Sorting procedures are a cornerstone methodology in asset pricing, because they seek to reveal if a particular feature helps understand cross-sectional differences between returns. While they are most often built according to particular firm characteristics (e.g., market capitalization, book-to-market ratio), they can also be processed with predictions so as to assess the accuracy of these forecasts.

The protocol we run is the following. For each month t , we train the models according to (6). From the resulting predictions $\hat{f}_t^M(\mathbf{x}_{t,i}, \mathcal{S}_t(\Delta T, D))$, we form Q sorted portfolios of homogeneous forecasts. These portfolios are held for a holding period Δh . Our dataset encompasses slightly more than 1,200 stocks, thus we compared $Q = 6$ groups of roughly 200 assets, which is enough to ensure a sufficient diversification of the portfolios. The first portfolio encompasses the stocks with the lowest predictions while the sixth gathers the ones with the highest forecasts. Within the portfolios, assets are weighted uniformly. Equally-weighted portfolios are indeed both simple and known to perform well in terms of Sharpe ratio, certainty-equivalent return, or turnover (DeMiguel, Garlappi, and Uppal (2009)). In Figure 7, we plot the average returns of the corresponding ranked portfolios.

Overall, the patterns are disappointing because one would expect the bars to increase with the portfolio ranks. While the top portfolio does outperform,¹² the bottom portfolios experience unexpectedly high returns, which they should not if the models were able to better discriminate

¹²This is not particularly due to exposure to very small stocks. The average market capitalization score of the top portfolios is 0.46, which indicates a mild small cap tilt. These portfolios also comprise 11%-12% of stocks in the bottom decile of market capitalization, hence tiny stocks are not significantly over-represented.

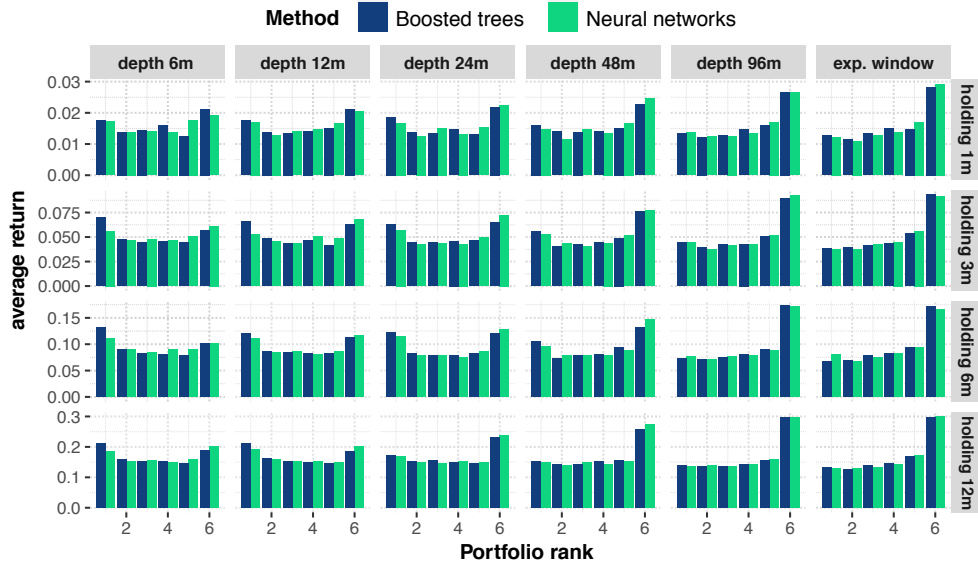


Fig. 7. **Average returns of long-only sorted portfolios.** We plot the out-of-sample mean returns of the 6 long-only ranked portfolios with uniform weights. Averages are computed across dates and labels. Rows of graphs corresponds to different holding periods while columns change with training sample depth. Trading occurs between January 2009 and February 2019. The ranges of averages depend on the holding periods. The latter are fixed for each row of the graph.

future low returns. The worst configuration is when the training sample is short (6 months) because the bars are almost flat, which highlights the inability of the models to correctly forecast future returns. A surprising outcome is the lack of difference between the two predictive models. Given the contrast in out-of-sample fit in Figure 5, it could have been expected that neural networks yielded better results. We postpone the study of this apparent contradiction to Section 4.2.

3.2. Long-short portfolios

The next step is to disentangle the effect of the three horizons: label maturity ΔT , training sample depth D and holding period Δh . To simplify the display of results, we compute the long-short returns between the two extreme portfolios (high (rank 6) minus low (rank 1) prediction). Since this is the only case for which we allow short positions, we will henceforth refer to it as the long-short portfolio. In Figure 8, we gather the average returns of these long-short strategies. The values are averaged over both model types, hence they can be viewed as the aggregation of an ensemble model for which each model would decide for half of the allocation.¹³ In short, we compute the vector of weights for each method, and average them. The corresponding cumulative returns are shown in Figure 9.

A first pattern we observe is that, overall, average returns increase with the training sample size: the more information the model sees, the more accurate its out-of-sample predictions turn out to be. For long-term labels (right portion of the figure), it seems that expanding windows add little value to the 8 year training samples. In all the combinations we tested, the shortest samples (from 6 to 24 months) are almost always associated to the lowest levels of average

¹³Strictly speaking, a pure ensemble method would aggregate the two predictions and allocate subsequently.

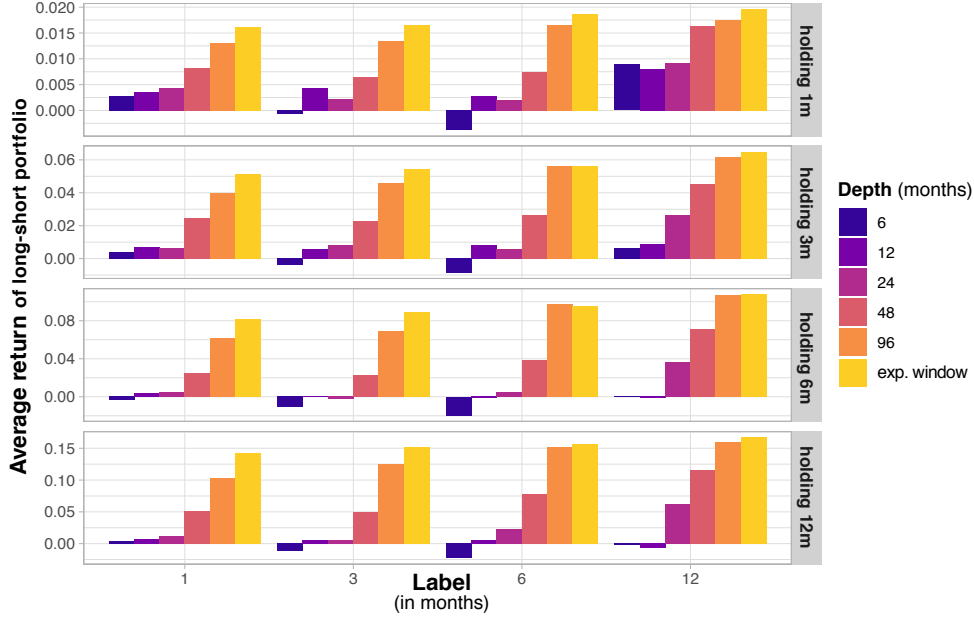


Fig. 8. **Average returns of long-short portfolios.** We plot the mean returns of the long-short (high minus low) portfolios. The average values aggregate the two vectors of weights (from the boosted trees and neural networks) in an ensemble fashion. Trading occurs between January 2009 and February 2019. The ranges of averages depend on the holding periods. The latter are fixed for each row of the graph.

returns. It seems that predictions based on small samples are too noisy to be competitive with estimates stemming from models trained on longer data samples.

The second finding from Figure 8 is that labels with longer maturities also generate more accurate forecasts. Except for two minor exceptions (12 month depths), the 12 month labels (on the right of the figure) always improve the results of the 1 month labels (on the left). This means that monthly rebalancing is more profitable when the predicted variable is the future *annual* return. In turn, this can be linked to Proposition 1, where it is shown that signal strength can be compounded and amplified across periods. However, all in all, the holding period does not seem to be very impactful: the rankings of bars do not change much from one line to another.

In a nutshell, our findings indicate that the most robust zone from an investment standpoint is the upper right quadrant, for large depth values. Short holding periods associated with long horizon labels seem like a winning combination in all our tests. However, frequent rebalancing is associated to higher asset rotation and increased trading fees.

3.3. Transaction costs and risk

While the above results are useful to assess the relative accuracy of the algorithms and methods, they are not sufficient for portfolio managers. To complement our analysis, we report the transaction cost-adjusted Sharpe ratio, which we compute as

$$SR^* = \frac{\bar{r} - 0.005 \times \text{Turn}}{\hat{\sigma}}, \quad \text{with} \quad \text{Turn} = \frac{1}{T-1} \sum_{t=2}^T \sum_{n=1}^{N_t} |w_t - w_{t-1}|. \quad (10)$$

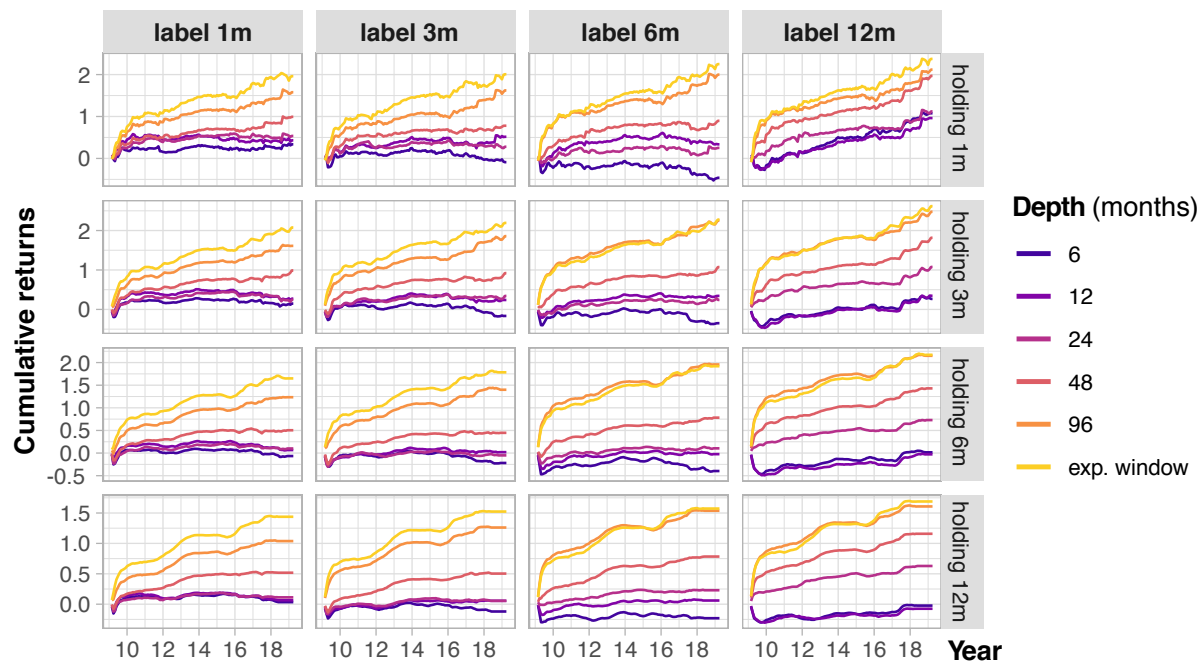


Fig. 9. **Cumulative returns.** We plot the cumulative returns the long-short (high minus low) portfolios defined in Section 3. Trading occurs between January 2009 and February 2019.

This indicator enhances the initial metric of raw returns in two directions. First, it takes risk into account by scaling average returns ($\bar{r}$) by their volatility ($\hat{\sigma}$), and second, it removes a proxy of transaction costs (the scaled turnover) in the numerator (the expression for Turn is taken from Brandt, Santa-Clara, and Valkanov (2009)). The average cost per unit of turnover (50 basis points) originates from Goto and Xu (2015) and is rather conservative. In the right equation of (10), N_t is the number of asset in the portfolio at time t .

The values of SR^* are gathered in Figure 10.¹⁴ Plainly, transaction costs do make a difference, and more indicators have negative values, compared to Figure 8. This is particularly true for short holding periods (which makes sense because asset rotation is more costly). Still, the best results are again obtained for deep samples and long-term samples, which further cements our conclusions.

The above empirical findings constitute the core of our argument. The models that perform the best are those that are trained with long-horizons labels over large data samples. An investor who rebalances every month or every quarter is better off when training models on annual returns, which may seem counter-intuitive. Our results show an area with low sensitivity: the yellow and orange bars of the bottom right graphs are all solidly positive. The combination of a six month holding period with a six month label and 96 month training sample is a reasonable choice because a shift in any direction does not alter the performance. This lack of sensitivity makes it a robust configuration from a modelling standpoint.

¹⁴The values of transaction cost adjusted average returns display fairly similar patterns because the volatility of portfolios is not too sensitive to depths and label horizons, though, in full disclosure, it increases mildly for long term labels.

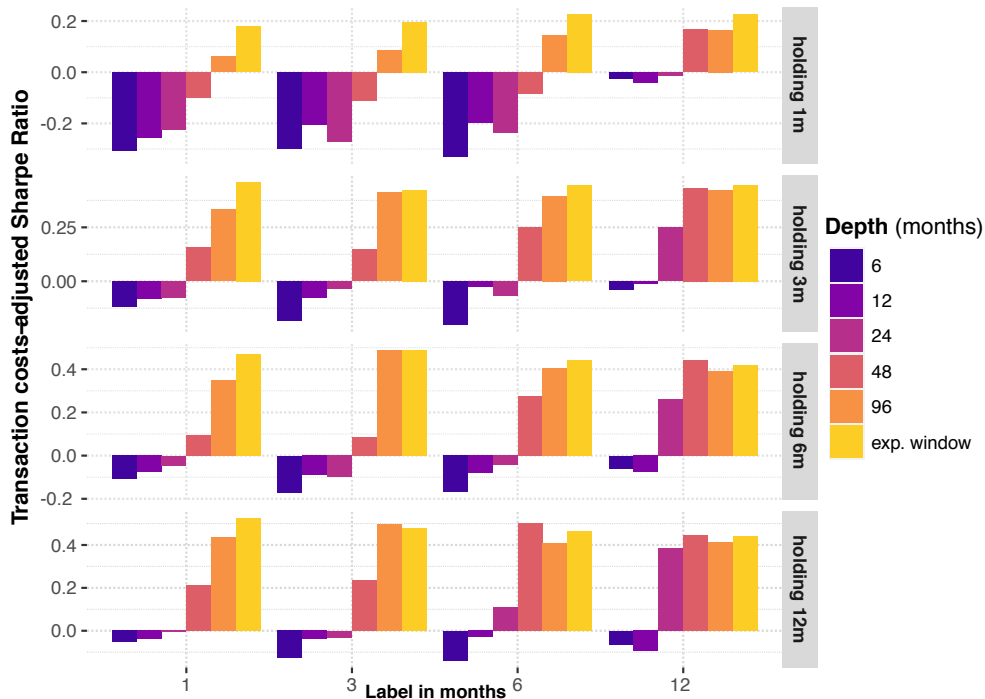


Fig. 10. **Transaction cost-adjusted Sharpe ratios.** We plot the metric defined in (10) for the long-short (high minus low) portfolios. The average values aggregate the two vectors of weights (from the boosted trees and neural networks) in an ensemble fashion. Trading occurs between January 2009 and February 2019. The ranges of metrics depend on the holding periods. The latter are fixed for each row of the graph.

4. Extensions

4.1. Feature persistence and variable importance

Because machine learning algorithms are black boxes, it is natural for investors to resort to techniques which reveal some of their properties. Methods of interpretability and explainability have recently blossomed and we refer to Molnar (2020) for a review on this topic. Technically, it is possible to characterize the behavior of models in the vicinity of each data point, but this is cumbersome to represent succinctly. Thus, for expositional clarity, it is customary to report aggregate views from so-called “global” interpretability models.

After each training round of the supervised models (i.e., each month), it is possible to track which features (characteristics) have the most impact. This can be done by evaluating the contribution of each variable in the reduction of the loss, or, equivalently, in the augmentation of the fit (measured as the sum of squared errors). We refer for instance to Section 4.4 in Molnar (2020) for tree-based algorithms in particular.

In Figure 11 below, we show the average variable importance across all training samples for two choices of sample depths (6 months versus 96 months) and two difference label horizons (1 month and 12 months). The metrics are normalized so that their maximum value is 100 in each subgraph. In three different configurations, trading volume (which is a proxy for liquidity) is the most important variable. Capital expenditures and price momentum are also often represented in the plots. In Gu et al. (2020), it is also found that momentum and liquidity are prominent

predictors in supervised models.

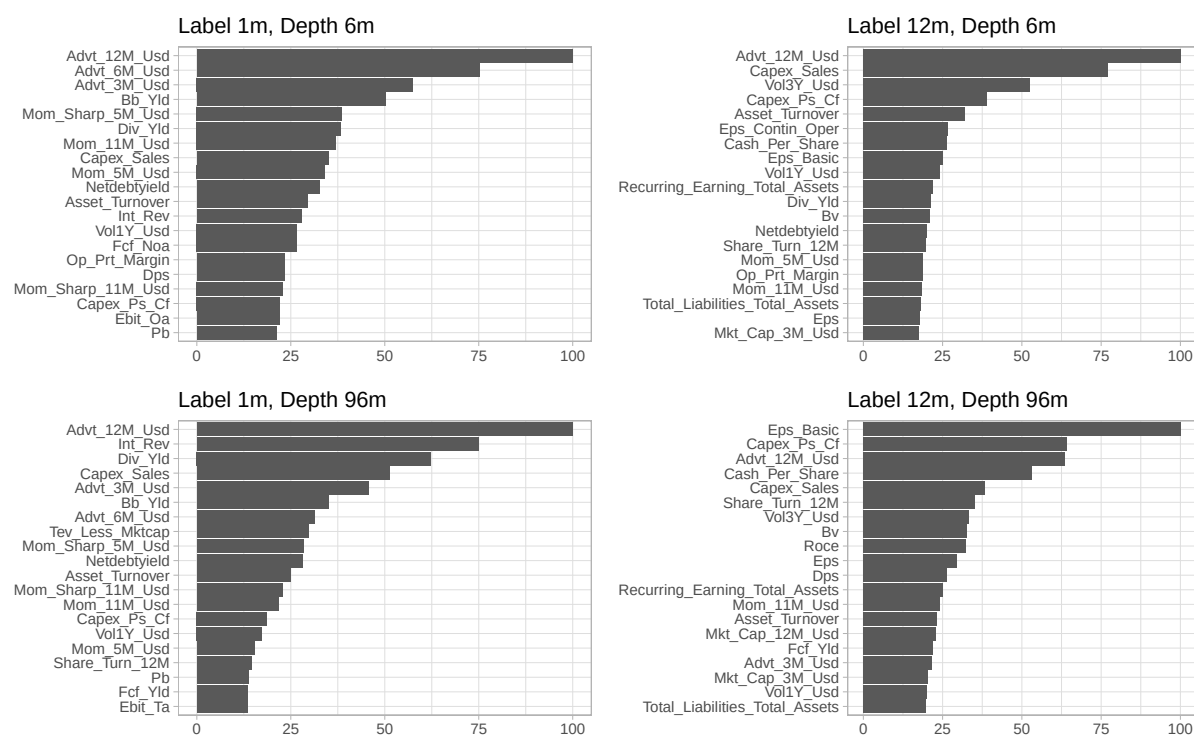


Fig. 11. **Average variable importance.** We plot the top 20 average variable importance metrics of boosted tree models trained between January 2009 and February 2019. Two sample depths are considered (6 months and 96 months), as well as two dependent variables (1 month forward and 12 month forward returns). For each choice of sample depth and label, variable importance is scaled to reach a maximum of 100.

There is nonetheless a clear difference between models trained on small samples with short term labels (upper left quadrant) and models trained on long samples with long term labels (lower right quadrant). For the former, liquidity is the most important factor, by far - with buyback, dividend yields and short-term momentum following (all three having a relatively low average autocorrelation: 0.52, 0.65 and 0.61, respectively - compared to a median of 0.84). For the latter, fundamental attributes, like earnings per share of capital expenditure ratios matter more (with average autocorrelation around 0.85).

In Figure 12, we display the relationship between variable importance and the mean autocorrelation of features (across stocks). To the left of 0.8 of autocorrelation, few points are available, which makes it unsuitable to draw conclusions. Between 0.8 and 0.92, the relationship is flat or mildly decreasing, and beyond 0.92 (where the mass of the points is), it is increasing. The slopes of local regression curves are more pronounced for models trained with short term labels. Notably, the rectangles locate the features with importance above 50. Apart for one exception (dividend yield), all such features are located to the right of the graph, beyond 0.88 of autocorrelation, meaning that they are highly persistent. The interested reader can find additional details on the most prominent features in Figure 15 in the Appendix.

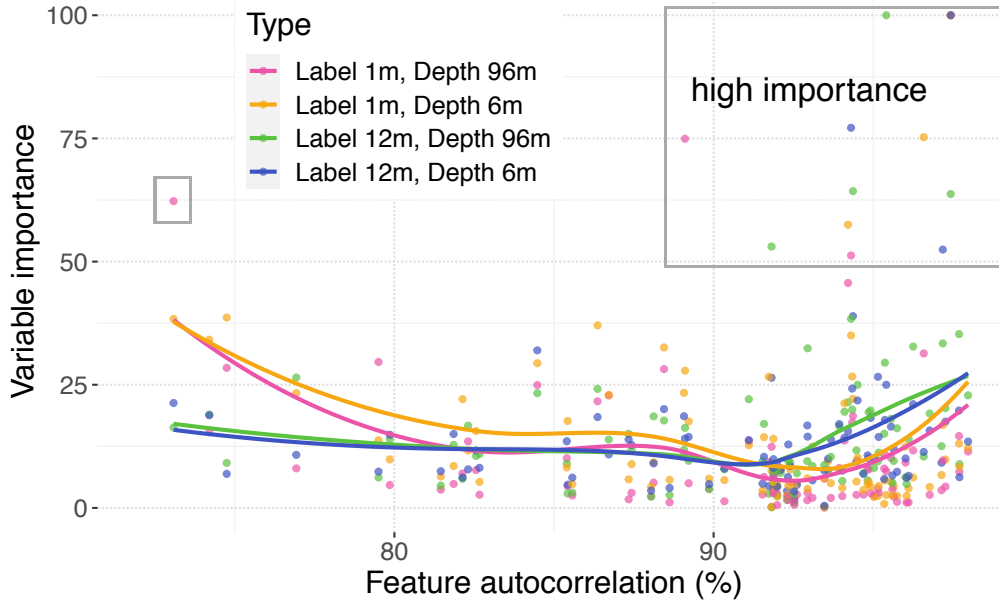


Fig. 12. **Relationship between feature persistence and variable importance.** We plot average variable importance as a function of average autocorrelation across stocks (see Tables 2 to 4 in the Appendix). The colored curves fit a local regression of loess type to show the trends. High values of variable importance are clustered in two grey rectangles. For each Type (choice of sample depth and label), variable importance is scaled to reach a maximum of 100.

4.2. Reconciling pure fit with portfolio performance

The usual loss decomposition in machine learning is the variance-bias tradeoff which shows that, as models gain accuracy on average (smaller bias), their dispersion increases (higher variance of forecasts) which in turn may augment the mean squared error. The net effect is often hard to handle and requires a fine tuning of the tradeoff between the two components. We do not follow this path in the present paper. Rather, we decompose the mean squared error into three simple components (we lighten the notations for simplicity):

$$\text{MSE} = (IT)^{-1} \sum_{t,i} (r_{t:t+\Delta T,i} - \hat{f}_t^M)^2 \quad (11)$$

$$\begin{aligned} &= (IT)^{-1} \left(\sum_{t,i} r_{t:t+\Delta T,i}^2 + \sum_{t,i} (\hat{f}_t^M)^2 - 2 \sum_{t,i} r_{t:t+\Delta T,i} \hat{f}_t^M \right) \\ &= \hat{\mathbb{V}}[r_{t:t+\Delta T,i}^2] + \hat{\mathbb{V}}[(\hat{f}_t^M)^2] - 2\hat{\text{Cov}}[r_{t:t+\Delta T,i}, \hat{f}_t^M], \end{aligned} \quad (12)$$

where we casually write $\hat{\mathbb{V}}[\cdot]$ and $\hat{\text{Cov}}[\cdot]$ for the sample variance and covariance functions. The above decomposition holds even if the summands $\hat{f}_t^M$ and $r_{t:t+\Delta T,i}$ do not have zero sums. The first term is irrelevant because model-independent. The second term evaluates the dispersion of predictions. Finally, the third term computes the cross relationship between the predicted and realized values. From an investment perspective, this last term is the most important. Indeed, it relates to the hit ratio which computes the frequency at which the sign of the return is correctly guessed. If the sign of the return is accurately predicted, the cross term is positive, which

contributes to the reduction of the squared error. When portfolios are constructed with predicted returns, the raw values and the fit of the predictions matter far less than their rankings.¹⁵ What is essential is that assets that will outperform have higher predictions, irrespectively of the level of the forecasts.

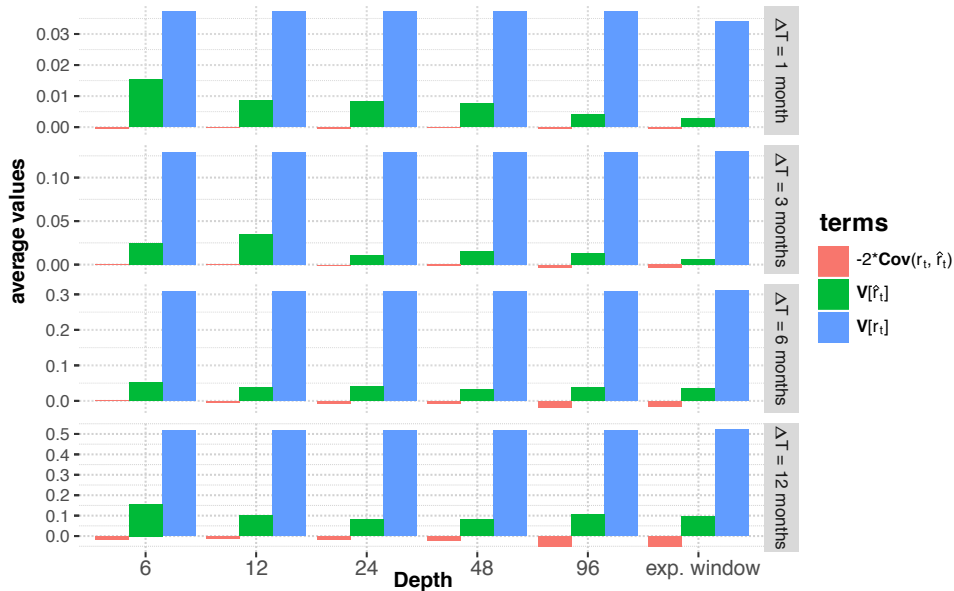


Fig. 13. **Decomposition of MSE.** We plot the average variance of realized returns, the average variance of predicted returns, and the average of minus two times the correlation between the realized and the predicted returns. At each date, we decompose the loss into the three terms identified in Equation (12). All values are then averaged over all dates and both model types. Each row of the graph pertains to one label horizon.

In Figure 13, we illustrate the average decomposition that we obtained with our predictions. For each depth, label, model type and point in time, we evaluate the three terms in (12) and subsequently average them across time and model types. Two salient patterns are remarkable. First, predictions are much less volatile, compared to realized returns. Second, the cross terms (covariances) account for only a very small portion of the MSE. These findings hold across any choice of label or sample depth. Nevertheless, the most favorable (i.e., negative) covariance terms are obtained for the deepest training sets (48 months and more) and for long horizon labels (6 and 12 months).

The dilution of covariances in the global fit is somewhat of a problem because of the marked connexion that these covariances have with realized returns. This relationship is plotted in Figure 14. The covariances are those depicted in Figure 13 (up to the negative scaling factor) while the long-short returns are those from Figure 8. Linear models fitted to capture the interaction between the two variables mostly yield significant slopes: only three of them out of 24 have a p -value above the 5% threshold. In comparison, the link between the variance of predictions and realized returns is very weak as only one p -value manages to pass below the 5% level.¹⁶

One major takeaway from these findings is that measures of cross-sectional fit between pre-

¹⁵In Wong, Chan, Azizi, and Xu (2020), the authors compute the correlation between the ranking of realized versus predicted returns. This idea is similar to what we propose in the present paper.

¹⁶Full results are available upon request.

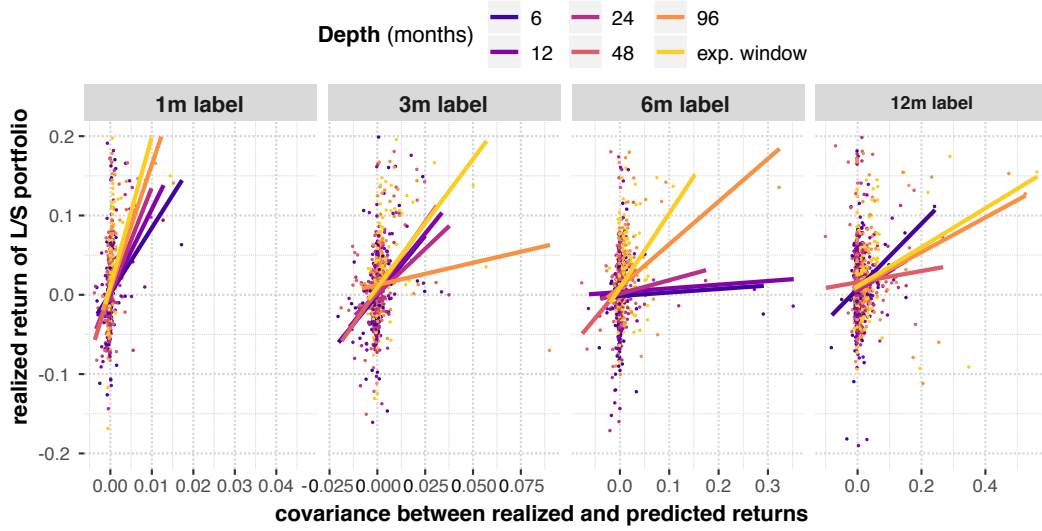


Fig. 14. **Relationship between covariances and average portfolio returns.** On the x -axis, we show the covariance between predicted returns and realized returns. The values are equal to $-1/2$ times those plotted in red in Figure 13. The y -axis shows the average realized return of the long-short strategy. Linear models are fitted for each sample depth. The largest p -value for the coefficient of the slope is 0.178 (obtained with 6 month labels). Note that the range of the x -axis depends on the horizon of the label.

dictions and realized values are much more important than squared errors and R^2 indicators that are commonly reported in the literature.

4.3. Learning with amnesic predictors

The analyses carried out above rely on a dataset with highly correlated features because sample autocorrelations have a median of 0.835. To further make the case of the importance of memory, it is useful to compare our results to models based on memoryless predictors. The simplest way to do so is to use the difference operator: $\Delta \mathbf{x}_{t,i} = \mathbf{x}_{t,i} - \mathbf{x}_{t-1,i}$. The learning process relying on $\Delta \mathbf{x}_{t,i}$ will focus on *variations* in feature values and not on feature levels. Using differences removes the memory in the time-series. The sample autocorrelations of features (stock-by-stock) are then concentrated around zero. More than 90% of them lie in the interval $[-0.3, 0.3]$. In Table 1, we report the mean monthly returns for the portfolios built on corresponding the ML signals. Similarly to our previous computations, we build 6 portfolios sorted on the predictions from the models and then average over the difference between the top portfolio and the bottom one.

The results speak unambiguously. Average returns based on variations of features are not only statistically insignificant, they are also often negative. For raw predictors, statistical significance is only reached for the largest sample depths, which is consistent with our previous conclusions. In addition, resorting to long term labels remains the best option to robustly increase average returns. Again, the most effective combination links autocorrelated features with persisting labels.

Label	Model	Depth					
		6 months	12 months	24 months	48 months	96 months	exp. window
PANEL A: models based on original features $\mathbf{x}_{t,i}$							
1 month	BT	0.004	0.004	0.003	0.007 (*)	0.013 (***)	0.015 (***)
1 month	NN	0.002	0.003	0.006	0.010 (***)	0.013 (***)	0.017 (***)
12 month	BT	0.012	0.012	0.011 (***)	0.012 (***)	0.017 (***)	0.020 (***)
12 month	NN	0.006	0.004	0.008 (*)	0.017 (***)	0.018 (***)	0.019 (***)
PANEL B: models based on amnesic features $\Delta\mathbf{x}_{t,i}$							
1 month	BT	-0.002	-0.003	-0.001	-0.002	0.000	0.000
1 month	NN	0.002	-0.002	-0.001	0.002	-0.002	0.001
12 month	BT	0.001	-0.000	-0.001	0.001	0.002	0.016 (*)
12 month	NN	-0.002	-0.000	0.000	0.000	0.000	0.011

Table 1: **Average monthly returns of ML-based portfolio strategies.** We report the mean returns of the long-short (high minus low) portfolios. Trading occurs between January 2009 and February 2019. Models use one month labels (low memory) or 12 month labels (high memory). The significance levels for the simple t -tests on the time-series of returns are: (***) $< 0.001 < (**) < 0.01 < (*) < 0.05$. BT stands for boosted trees and NN for neural networks.

5. Conclusion

In this paper, we study the impact of the autocorrelation of dependent (output) and independent (input) variables in the out-of-sample accuracy of factor-based machine learning models. While autocorrelation is usually inconvenient in classical econometric analysis because it perturbs inference, we find that it is in fact valuable from a pure predictive standpoint. Theoretically, it is possible to show that memory in underlying processes can diffuse through time and compound the signal contained in the predictors.

When estimating rolling machine learning models every month, we find that these models perform better when the sample size is large and when the dependent variable (stock return) is computed over a long horizon. Empirically, we find discrepancies in the relative importance of features in the trained models. Some categories, e.g., liquidity, momentum and earnings, seem to play a prominent role, but this role differs depending on the horizon of predicted returns.

These conclusions are useful for anyone willing to build predictive models for the cross-section of stock returns based on firm-specific characteristics. The main applications are naturally portfolio choice and empirical asset pricing. Notably, our results remain robust when including trading costs and when accounting for risk.

6. Acknowledgments

The authors are grateful for the comments of three anonymous referees which have helped improve the expositional quality of the paper.

Appendix A. Data

Predictor	Description	Avg. autocorr.
Advt_12M_Usd	avg. daily volume in amount in USD over 12 months	0.88
Advt_3M_Usd	avg. daily volume in amount in USD over 3 months	0.83
Advt_6M_Usd	avg. daily volume in amount in USD over 6 months	0.86
Asset_Turnover	total sales on average assets	0.78
Bb_Yld	buyback yield	0.52
Bv	book value	0.97
Capex_Ps_Cf	capital expenditure on price to sale cash flow	0.85
Capex_Sales	capital expenditure on sales	0.83
Cash_Div_Cf	cash dividends cash flow	0.74
Cash_Per_Share	cash per share	0.82
Cf_Sales	cash flow per share	0.84
DebtEquity	debt to equity	0.85
Div_Yld	dividend yield	0.65
Dps	dividend per share	0.70
Ebit_Bv	EBIT on book value	0.74
Ebit_Noa	EBIT on non operating asset	0.72
Ebit_Oa	EBIT on operating asset	0.73
Ebit-Ta	EBIT on total asset	0.73
Ebitda_Margin	EBITDA margin	0.74
Eps	earnings per share	0.85
Eps_Basic	earnings per share basic	0.85
Eps_Basic_Gr	earnings per share growth	0.78
Eps_Contin_Oper	earnings per share continuing operations	0.85
Eps_Dil	earnings per share diluted	0.85
Ev	enterprise value	0.87
Ev_Ebitda	enterprise value on EBITDA	0.71
Fa_Ci	fixed assets on common equity	0.87
Fcf	free cash flow	0.82
Fcf_Bv	free cash flow on book value	0.82
Fcf_Ce	free cash flow on capital employed	0.76
Fcf_Margin	free cash flow margin	0.81
Fcf_Noa	free cash flow on net operating assets	0.81
Fcf_Oa	free cash flow on operating assets	0.82
Fcf-Ta	free cash flow on total assets	0.82
Fcf_Tbv	free cash flow on tangible book value	0.79
Fcf-Toa	free cash flow on total operating assets	0.79
Fcf_Yld	free cash flow yield	0.80

Table 2: **List of predictors.** We list the features used in our study, along with their average autocorrelation (across all stocks).

Predictor	Description	Avg. autocorr.
Free_Ps_Cf	free cash flow on price sales	0.82
Int_Rev	intangibles on revenues	0.81
Interest_Expense	interest expense coverage	0.86
Mkt_Cap_12M_Usd	average market cap. over 12 months in USD	0.89
Mkt_Cap_3M_Usd	average market cap. over 3 months in USD	0.87
Mkt_Cap_6M_Usd	average market cap. over 6 months in USD	0.88
Mom_11M_Usd	momentum 12 - 1 months in USD	0.72
Mom_5M_Usd	momentum 6 - 1 months in USD	0.60
Mom_Sharp_11M_Usd	momentum 12 - 1 months in USD divided by vol.	0.73
Mom_Sharp_5M_Usd	momentum 6 - 1 months in USD divided by vol.	0.61
Nd_Ebitda	net debt on EBITDA	0.77
Net_Debt	net debt	0.86
Net_Debt_Cf	net debt on cash flow	0.78
Net_Margin	net margin	0.84
Netdebtyield	net debt yield	0.78
Ni	net income	0.85
Ni_Avail_Margin	net income available margin	0.83
Ni_Oa	net income on operating asset	0.84
Ni_Toa	net income on total operating asset	0.81
Noa	net operating asset	0.90
Oa	operating asset	0.91
Ocf	operating cash flow	0.84
Ocf_Bv	operating cash flow on book value	0.83
Ocf_Ce	operating cash flow on capital employed	0.77
Ocf_Margin	operating cash flow margin	0.82
Ocf_Noa	operating cash flow on net operating assets	0.82
Ocf_Oa	operating cash flow on operating assets	0.82
Ocf-Ta	operating cash flow on total assets	0.82
Ocf_Tbv	operating cash flow on tangible book value	0.80
Ocf_Toa	operating cash flow on total operating assets	0.79
Op_Margin	operating margin	0.85
Op_Prt_Margin	net margin 1Y growth	0.78
Oper_Ps_Net_Cf	cash flow from operations per share net	0.83
Pb	price to book	0.85
Pe	price earnings	0.81
Ptx_Mgn	margin pretax	0.85
Recurring_Earning_Total_Assets	recurring earnings on total assets	0.87
Return_On_Capital	return on capital	0.83

Table 3: **List of predictors (continued)**. We list the features used in our study, along with their average autocorrelation (across all stocks).

Predictor	Description	Avg. autocorr.
Rev	revenue	0.90
Roa	return on assets	0.83
Roc	return on capital	0.83
Roce	return on capital employed	0.83
Roe	return on equity	0.83
Sales_Ps	price to sales	0.88
Share_Turn_12M	average share turnover 12 months	0.86
Share_Turn_3M	average share turnover 3 months	0.80
Share_Turn_6M	average share turnover 6 months	0.84
Ta	total assets	0.92
Tev_Less_Mktcap	total enterprise value less market capitalization	0.71
Tot_Debt_Rev	total debt on revenue	0.83
Total_Capital	total capital	0.92
Total_Debt	total debt	0.84
Total_Debt_Capital	total debt on capital	0.86
Total_Liabilities_Total_Assets	total liabilities on total assets	0.88
Vol1Y_Usd	volatility of returns over one year	0.82
Vol3Y_Usd	volatility of returns over 3 years	0.87

Table 4: **List of predictors (continued)**. We list the features used in our study, along with their average autocorrelation (across all stocks).

Appendix B. Persistence and variable importance

Feature	Label 1m, Depth 96m	Label 12m, Depth 96m	Label 1m, Depth 6m	Label 12m, Depth 6m	Autocorrelation	Total importance
Advt_12M_Usd	100.0	63.7	100.0	100.0	97.4	363.7
Capex_Sales	51.2	38.4	35.0	77.2	94.3	201.8
Capex_Ps_Cf	18.7	64.3	22.1	38.9	94.4	144.0
Advt_3M_Usd	45.7	21.5	57.5	15.5	94.2	140.1
Div_Yld	62.3	16.3	38.4	21.3	73.1	138.2
Eps_Basic	7.4	100.0	2.6	25.0	95.4	135.1
Int_Rev	74.9	16.3	27.8	14.1	89.1	133.1
Advt_6M_Usd	31.4	10.4	75.2	11.4	96.6	128.4
Vol3Y_Usd	9.6	33.4	16.8	52.4	97.2	112.3
Asset_Turnover	24.9	23.3	29.4	32.0	84.5	109.6
Mom_11M_Usd	21.6	24.2	37.1	18.4	86.4	101.3
Cash_Per_Share	12.4	53.1	7.5	26.4	91.8	99.4
Netdebtlyield	28.2	17.7	32.6	20.1	88.4	98.5
Vol1Y_Usd	17.4	20.0	26.7	24.2	94.3	88.2
Mom_5M_Usd	15.3	18.9	34.1	18.8	74.2	87.2
Mom_Sharp_5M_Usd	28.4	9.1	38.7	6.9	74.7	83.1
Share_Turn_12M	14.6	35.3	13.1	19.8	97.7	82.7
Bv	11.7	32.8	6.2	21.1	96.3	71.7
Mom_Sharp_11M_Usd	22.8	13.9	22.9	10.9	86.7	70.5
Pb	13.7	19.5	21.2	14.3	94.1	68.7
Dps	8.0	26.5	23.3	10.8	76.9	68.6
Recurring_Earning_Total_Assets	9.6	25.0	10.2	21.8	94.5	66.6
Op_Prt_Margin	10.7	11.8	23.3	18.6	89.1	64.4
Eps	10.8	29.5	5.6	17.9	95.4	63.7
Fcf_Yld	13.5	22.1	12.7	13.8	91.1	62.1
Mkt_Cap_12M_Usd	11.4	22.9	12.2	13.5	98.0	59.9
Tev_Less_Mktcap	29.6	6.2	13.7	7.4	79.5	56.9
Roce	3.1	32.4	3.9	14.5	92.9	53.8
Mkt_Cap_3M_Usd	4.3	20.2	9.8	17.5	97.3	51.9
Interest_Expense	10.2	18.1	5.3	16.5	95.6	50.0
Ebit-Ta	13.5	16.7	11.6	7.8	82.3	49.7
Fa_Ci	8.1	16.6	7.6	17.0	93.9	49.3
Share_Turn_3M	6.3	17.1	14.4	10.4	91.6	48.1
Eps_Basic_Gr	5.0	9.6	17.6	14.4	89.2	46.6
Cash_Div_Cf	7.7	10.5	15.6	11.7	82.6	45.5
Total_Liabilities_Total_Assets	3.0	19.8	4.3	18.3	94.9	45.3
Ev_Ebitda	4.7	14.0	9.8	14.9	79.9	43.3
Fcf_NoA	9.3	2.9	26.6	4.4	91.7	43.3
Ni	7.8	19.4	5.2	10.1	94.9	42.6
Eps_Contin_Oper	3.9	9.5	2.4	26.6	95.1	42.4
Net_Debt_Cf	5.1	18.6	9.3	9.0	88.1	42.0
Sales_Ps	2.7	19.2	4.3	15.7	96.8	41.8
Ebit_Bv	4.9	12.8	8.5	15.0	81.9	41.2
Ebit_Oa	7.0	5.9	22.1	6.2	82.1	41.2
Nd_Ebitda	10.1	9.0	8.2	13.5	85.4	40.8
Share_Turn_6M	9.2	6.2	12.5	10.0	96.0	37.9
Ocf_Tbv	3.1	11.9	8.9	12.8	87.4	36.6
Fcf_Tbv	1.7	15.0	5.8	13.9	87.3	36.5
Ocf_Margin	6.6	5.4	14.0	9.9	91.9	35.9
Cf_Sales	7.7	7.7	9.5	10.0	93.6	34.8

Fig. 15. **Relationship between feature persistence and variable importance.** We show average variable importance as a function of average autocorrelation across stocks (see Tables 2 to 4 in the Appendix). High values of variable importance are clustered in two grey rectangles. For each Type (choice of sample depth and label), variable importance is scaled to reach a maximum of 100. Total importance is the sum across the 4 configurations.

Appendix C. Proof of Proposition 1

We have:

$$\begin{aligned} r_* &= (1 + r_{t+1})(1 + r_{t+2}) - 1 \\ &= (1 + \beta C_t + \epsilon_{t+1}^r)(1 + \beta C_{t+1} + \epsilon_{t+2}^r) - 1 \\ &= \beta C_t + \beta C_{t+1} + \beta^2 C_t C_{t+1} \end{aligned} \quad (13)$$

$$+ \epsilon_{t+1}^r + \epsilon_{t+2}^r + \epsilon_{t+1}^r \epsilon_{t+2}^r + \beta \epsilon_{t+1}^r C_{t+1} + \beta \epsilon_{t+2}^r C_t, \quad (14)$$

where the two lines (13) and (14) are organized so as to mark the difference with pure signal (13) terms and noisy terms (14). The last two terms could be considered as hybrid, but we take a conservative stance and classify them as noise. The SN ratio of r_* is thus equal to

$$\text{SN}(r_*) = \frac{\mathbb{V}[\beta C_t + \beta C_{t+1} + \beta^2 C_t C_{t+1}]}{\mathbb{V}[\epsilon_{t+1}^r + \epsilon_{t+2}^r + \epsilon_{t+1}^r \epsilon_{t+2}^r + \beta \epsilon_{t+1}^r C_{t+1} + \beta \epsilon_{t+2}^r C_t]}. \quad (15)$$

The numerator of (15) is equal to

$$\begin{aligned} \text{num} &= \mathbb{V}[\beta C_t + \beta C_{t+1} + \beta^2 C_t C_{t+1}] \\ &= 2\beta^2 \sigma_C^2 (1 - \rho_C^2)^{-1} + \beta^4 \mathbb{V}[C_t C_{t+1}] \\ &\quad + 2(\beta^2 \text{cov}(C_t, C_{t+1}) + \beta^3 \text{cov}(C_t, C_t C_{t+1}) + \beta^3 \text{cov}(C_{t+1}, C_t C_{t+1})) \end{aligned}$$

Because of the normality of the white noises, the couple (C_t, C_{t+1}) follows a bivariate Gaussian distribution with correlation equal to ρ_C . The marginal laws are identical with zero mean and $\frac{\sigma_C^2}{1-\rho_C^2}$ variance. Moments of bivariate Gaussian distributions are well known (see Section 46.3 in [Kotz, Balakrishnan, and Johnson \(2004\)](#)). Odd moments (when the sum of powers is odd) are equal to zero. Moreover,

$$\mathbb{V}[C_t C_{t+1}] = \mathbb{E}[(C_t C_{t+1})^2] - \mathbb{E}[C_t C_{t+1}]^2 = (1 + 2\rho_C^2) \frac{\sigma_C^4}{(1 - \rho_C^2)^2} - \rho_C^2 \frac{\sigma_C^4}{(1 - \rho_C^2)^2},$$

where the first term comes from (46.23) in [Kotz et al. \(2004\)](#).

Hence,

$$\text{num} = 2\beta^2 \sigma_C^2 \left(\frac{1 + \rho_C}{1 - \rho_C^2} \right) + \beta^4 \frac{\sigma_C^4}{(1 - \rho_C^2)^2} (1 + \rho_C^2).$$

Taking into account the independence between ϵ^r and C and between ϵ_s^r and ϵ_t^r ($s \neq t$), the denominator of (15) is only defined by the variance of the individual terms (all cross covariance terms vanish):

$$\text{den} = \mathbb{V}[\epsilon_{t+1}^r + \epsilon_{t+2}^r + \epsilon_{t+1}^r \epsilon_{t+2}^r + \beta \epsilon_{t+1}^r C_{t+1} + \beta \epsilon_{t+2}^r C_t] \quad (16)$$

$$= 2\sigma_r^2 + \mathbb{V}[\epsilon_{t+1}^r \epsilon_{t+2}^r] + \beta^2 \mathbb{V}[\epsilon_{t+1}^r] \mathbb{V}[C_{t+1}] + \beta^2 \mathbb{V}[\epsilon_{t+2}^r] \mathbb{V}[C_t] \quad (17)$$

$$= 2\sigma_r^2 + \sigma_r^4 + 2\beta^2 \sigma_r^2 \sigma_C^2 (1 - \rho_C^2)^{-1}, \quad (18)$$

where we also used the fact that the processes C and ϵ^r have zero means. The SN ratio of r_* is

larger than that of r_t when

$$\begin{aligned}
\text{SN}(r_t) = \beta^2 \frac{\sigma_C^2}{\sigma_r^2(1-\rho_C^2)} &\leq \frac{2\beta^2\sigma_C^2 \left(\frac{1+\rho_C}{1-\rho_C^2}\right) + \beta^4 \frac{\sigma_C^4}{(1-\rho_C^2)^2} (1+\rho_C^2)}{2\sigma_r^2 + \sigma_r^4 + 2\beta^2\sigma_r^2\sigma_C^2(1-\rho_C^2)^{-1}} = \text{SN}(r_*) \\
\Leftrightarrow \quad 1 &\leq \frac{2(1+\rho_C) + \beta^2 \frac{\sigma_C^2}{1-\rho_C^2} (1+\rho_C^2)}{2 + \sigma_r^2 + 2\beta^2\sigma_C^2(1-\rho_C^2)^{-1}} \\
\Leftrightarrow \quad \sigma_r^2 + 2\beta^2\sigma_C^2(1-\rho_C^2)^{-1} &\leq 2\rho_C + \beta^2 \frac{\sigma_C^2}{1-\rho_C^2} (1+\rho_C^2) \\
\Leftrightarrow \quad \sigma_r^2 &\leq 2\rho_C - \beta^2\sigma_C^2,
\end{aligned}$$

which is the sought result.

References

- Aldridge, I. and M. Avellaneda (2019). Neural networks in finance: Design and performance. *Journal of Financial Data Science* 1(4), 39–62.
- Amihud, Y. and C. M. Hurvich (2004). Predictive regressions: A reduced-bias estimation method. *Journal of Financial and Quantitative Analysis* 39(4), 813–841.
- Arjovsky, M., L. Bottou, I. Gulrajani, and D. Lopez-Paz (2019). Invariant risk minimization. *arXiv Preprint* (1907.02893).
- Bandi, F. M., B. Perron, A. Tamoni, and C. Tebaldi (2019). The scale of predictability. *Journal of Econometrics* 208(1), 120–140.
- Bartlett, P. L. (1998). The sample complexity of pattern classification with neural networks: the size of the weights is more important than the size of the network. *IEEE Transactions on Information Theory* 44(2), 525–536.
- Baum, E. B. and D. Haussler (1989). What size net gives valid generalization? *Neural Computation* 1(1), 151–160.
- Boudoukh, J., R. Israel, and M. P. Richardson (2020). Biases in long-horizon predictive regressions. *SSRN Working Paper* 3612600.
- Boudoukh, J., M. Richardson, and R. F. Whitelaw (2008). The myth of long-horizon predictability. *Review of Financial Studies* 21(4), 1577–1605.
- Brandt, M. W., P. Santa-Clara, and R. Valkanov (2009). Parametric portfolio policies: Exploiting characteristics in the cross-section of equity returns. *Review of Financial Studies* 22(9), 3411–3447.
- Campbell, J. Y. (2001). Why long horizons? a study of power against persistent alternatives. *Journal of Empirical Finance* 8(5), 459–491.
- Campbell, J. Y., A. W. Lo, and A. C. MacKinlay (1997). *The econometrics of financial markets*. Princeton University Press.
- Campbell, J. Y. and M. Yogo (2006). Efficient tests of stock return predictability. *Journal of Financial Economics* 81(1), 27–60.
- Chen, L., M. Pelger, and J. Zhu (2019). Deep learning in asset pricing. *SSRN Working Paper* 3350138.
- Chen, T. and C. Guestrin (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International conference on knowledge discovery and data mining*, pp. 785–794. ACM.
- Chen, W. W. and R. S. Deo (2009). Bias reduction and likelihood-based almost exactly sized hypothesis testing in predictive regressions using the restricted likelihood. *Econometric Theory* 25(5), 1143–1179.
- Chevillon, G. (2017). Robustness of multistep forecasts and predictive regressions at intermediate and long horizons. *SSRN Working Paper* 3019901.

- Coqueret, G. and R. Deguest (2020). Predictive regressions: A machine learning perspective. *SSRN Working Paper 3709412*.
- Coqueret, G. and T. Guida (2020). *Machine Learning for Factor Investing*. www.mlfactor.com: CRC Press.
- DeMiguel, V., L. Garlappi, and R. Uppal (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies* 22(5), 1915–1953.
- Diebold, F. and R. Mariano (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics* 13(3), 253–265.
- Du, K.-L. and M. N. Swamy (2013). *Neural networks and statistical learning*. Springer Science & Business Media.
- Fama, E. F. and K. R. French (1988a). Dividend yields and expected stock returns. *Journal of Financial Economics* 22(1), 3–25.
- Fama, E. F. and K. R. French (1988b). Permanent and temporary components of stock prices. *Journal of Political Economy* 96(2), 246–273.
- Feng, G., N. Polson, and J. Xu (2019). Deep learning in characteristics-sorted factor models. *SSRN Working Paper 3243683*.
- Freyberger, J., A. Neuhierl, and M. Weber (2020). Dissecting characteristics nonparametrically. *Review of Financial Studies* 33(5).
- Gonzalo, J. and J.-Y. Pitarakis (2018). Predictive regressions. In *Oxford Research Encyclopedia of Economics and Finance*.
- Goto, S. and Y. Xu (2015). Improving mean variance optimization through sparse hedging restrictions. *Journal of Financial and Quantitative Analysis* 50(6), 1415–1441.
- Gu, S., B. T. Kelly, and D. Xiu (2020). Empirical asset pricing via machine learning. *Review of Financial Studies* 33(5), 2223–2273.
- Gu, S., B. T. Kelly, and D. Xiu (2021). Autoencoder asset pricing models. *Journal of Econometrics* 222(1), 429–450.
- Hjalmarsson, E. (2008a). Interpreting long-horizon estimates in predictive regressions. *Finance Research Letters* 5(2), 104–117.
- Hjalmarsson, E. (2008b). The Stambaugh bias in panel predictive regressions. *Finance Research Letters* 5(1), 47–58.
- Hjalmarsson, E. (2011). New methods for inference in long-horizon regressions. *Journal of Financial and Quantitative Analysis* 46(3), 815–839.
- Kelly, B. T., S. Pruitt, and Y. Su (2019). Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics* 134(3), 501–524.
- Koijen, R. S. and M. Yogo (2019). A demand system approach to asset pricing. *Journal of Political Economy* 127(4), 1475–1515.

- Kotz, S., N. Balakrishnan, and N. L. Johnson (2004). *Continuous multivariate distributions, Volume 1: Models and applications (Second Edition)*, Volume 1.
- Krauss, C., X. A. Do, and N. Huck (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research* 259(2), 689–702.
- Lawrence, S., C. L. Giles, and A. C. Tsoi (1998). What size neural network gives optimal generalization? Convergence properties of backpropagation. Technical report.
- Lettau, M. and M. Pelger (2020a). Estimating latent asset-pricing factors. *Journal of Econometrics* 218(1), 1–31.
- Lettau, M. and M. Pelger (2020b). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies* 33(5), 2274–2325.
- Molnar, C. (2020). *Interpretable Machine Learning*. <https://christophm.github.io/interpretable-ml-book/>: Lulu.
- Phillips, P. C. and J. H. Lee (2013). Predictive regression under various degrees of persistence and robust long-horizon regression. *Journal of Econometrics* 177(2), 250–264.
- Sizova, N. (2014). A frequency-domain alternative to long-horizon regressions with application to return predictability. *Journal of Empirical Finance* 28, 261–272.
- Stambaugh, R. F. (1999). Predictive regressions. *Journal of Financial Economics* 54(3), 375–421.
- Valkanov, R. (2003). Long-horizon regressions: theoretical results and applications. *Journal of Financial Economics* 68(2), 201–232.
- Wong, S. Y., J. Chan, L. Azizi, and R. Y. Xu (2020). Time-varying neural network for stock return prediction. *arXiv Preprint* (2003.02515).
- Xu, K.-L. (2020). Testing for multiple-horizon predictability: Direct regression based versus implication based. *Review of Financial Studies* 33(9), 4403–4443.